

Emmet McEvilly

emcewill@andrew.cmu.edu — (424) 383-0492 — emcev.com

EDUCATION

Carnegie Mellon University, Pittsburgh, PA

Bachelor of Science in Mechanical Engineering

Expected May 2028

GPA: 4.0 — Dean's List

Phillips Academy Andover, Andover, MA — High School Diploma

June 2024

EXPERIENCE & PROJECTS

Mechanical Design Engineer, MetaMobility Lab (CMU)

May 2026 – Present

- Engineered a lightweight, torque-assist ankle exoskeleton to support gait stability and rehabilitation research
- Designed mechanical components and interfaces in SolidWorks; 3D printed rapid prototypes to iterate on fit, strength, and functionality
- Executed bench and human-subject testing across 5 sessions to evaluate fit, comfort, torque delivery, and performance
- Implemented gait-phase-based control logic and integrated the actuator into the lab's closed-loop torque control system
- Documented the hardware design, CAD, and manufacturing process as an open-source platform for other labs to reproduce

Co-Founder & Product Engineer, Tilsit

Jun 2025 – Jun 2026

- Co-founded a community-based social prediction app where users make micro-predictions on real-world events and compete via an ELO-style rating system
- Facilitated 10 user interviews and market research to identify user needs and refine product direction
- Directed visual and product design in Figma, developing wireframes, user flows, and interaction patterns
- Built and deployed full-stack functionality using React Native, Supabase, and AWS, including database architecture and Row Level Security (RLS) policies, scaling to 500 active users

Propulsion Team Member, Carnegie Mellon Solar Racing

Jan 2026 – Present

- Designed and developed a motor housing system for the team's solar race boat
- Assembled and tested components, ensuring alignment, durability, and thermal management
- Performed 3D scanning to capture existing system geometry for digital documentation and future design reference

Physics 1 Course Center (CMU) — Teaching Assistant Associate

- Guided 50 students per semester by answering questions, reviewing concepts, and coaching problem-solving strategies

Innovation, Design, and Entrepreneurship (CMU) — Student Project

- Spearheaded market research and stakeholder interviews across the biking industry, including CMU's bike infrastructure consultant, to develop an MVP concept addressing biker safety

Fundamentals of Mechanical Engineering (CMU) — Student Project

- Applied CAD modeling, 3D printing, and fluid simulations across team-based projects

LEADERSHIP & ACTIVITIES

Belvedere Tennis Club — Tennis Instructor

- Led youth tennis camps and private lessons for 20+ students per season, engaging all ages and skill levels while fostering a collaborative environment
- Built communication and leadership skills transferable to team-based engineering projects

Tamalpais High School — Varsity Tennis Captain

- Captained a 12-player roster to league finals, running morning practices and driving team cohesion and development

SKILLS

Design & Fabrication: SolidWorks, CAD, 3D Printing, Rapid Prototyping

Programming & Controls: Python, MATLAB, Java, Closed-Loop Control, Embedded Systems

Other Tools: Figma, Adobe Suite, AWS, Supabase